Total No. of Questions : 6]

SEAT No. :

PD81

[Total No. of Pages : 2

[6410]-402

T.E. (Civil) (Insem)

DESIGN OF REINFORCED CONCRETE STRUCTURES

(2019 Pattern) (Semester - II) (301013)

Time : 1¼ Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
2) *Figures to the right indicate full marks.*
3) *Use of IS 456-2000 and non programmable calculator is allowed.*
4) *Neat diagrams must be drawn wherever necessary.*
5) *Mere reproduction from IS Code as answer, will not be given full credit.*
6) *Assume any other data, if necessary.*

Q1) a) Explain balanced and under reinforced section with suitable sketch. **[4]**

b) A beam of size 230mm × 550mm effective depth is simply supported over a span of 5m. The reinforcement consists of 5 bars of 20mm diameter at tension face. Find intensity of uniformly distributed load (including self-weight) that can be carried by beam. Use M25 & Fe415. **[6]**

OR

Q2) a) Explain the terms bond stress and development length. Calculate development length for 16mm diameter bar of grade Fe500 and M30 grade of concrete in tension and compression using LSM. **[4]**

b) Calculate maximum safe superimposed load carried by beam of effective span 8m. The beam is detailed as below- **[6]**

i) Width of rib = 300mm

ii) Effective flange width = 1200mm

iii) Thickness of flange = 130mm

iv) Effective depth = 565mm

v) Tension steel = 5 nos. of 25mm diameter

vi) M25 grade of concrete and Fe500 grade of steel

vii) Effective cover = 40mm

Q3) a) Explain four reason where doubly reinforced section is required. **[2]**

b) Design cantilever slab using LSM approach for an effective span of 1.8m carrying live load of 2.5kN/m^2 and floor finish of 1.5kN/m^2. Use M20 & Fe 415. Draw the details of the reinforcement. **[8]**

OR

Q4) a) What is the purpose of partial safety factors used in LSM? Give the partial safety factors for stresses in steel and concrete. **[2]**

b) A RC slab is to be provided for a passage measuring 8.5m×2.5m with 230mm wide beams around all edges. Design the suitable slab assuming LL 4kN/m^2 and FF 1.0kN/m^2. Use M30 and Fe500. Assume moderate exposure condition. Show details of the reinforcement. **[8]**

Q5) Design a simply supported two way slab of effective spans 5.5m×4.5m effective carrying L.L. of 3kN/m^2 and F.F of 1.5 kN/m^2. Use M25 and Fe415 for mild exposure condition. Draw the details of the reinforcement. (Neglect design of distribution steel and check for shear) **[10]**

OR

Q6) Design a two way slab of effective spans 4.5m×3.5m with two adjacent edges discontinuous. The slab is supported on beams of 300m width around all edges. Provide L.L of 3kN/m^2 and F.F. of 1kN/m^2. Use M20 and Fe415. Show the details of the reinforcement with neat and clean sketch. (Neglect design of distribution steel, torsion reinforcement and check for shear) **[10]**

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